Bounded Cognitive Diagnosis Models

Beta Responses, Latent Attribute Profiles, and Bayesian Classification

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1. Introduction: Motivation and Scope

- Cognitive diagnosis models describe units using discrete latent attributes.
- Bounded continuous indicators retain information that binary cutoffs discard.
- Beta B-DINA links indicators to attribute profiles through a conjunctive Q-matrix gate (Oliveira et al. 2026a).
- The application uses municipal Economy, Health, and Education attributes.
- The companion chapter is `BoundedCDM.Rmd`.

1.1 Bounded Responses and Diagnostic Questions

Examples include visual analogue scales, proportions, and development indices.

For theoretical bounds $L < U$, use

$$Y_{ij} = \frac{Y'_{ij} - L}{U - L}.$$

- The beta observation model requires $0 < Y_{ij} < 1$.
- Observed responses are continuous. Latent attributes are binary.
- Diagnosis asks which combination of attributes best describes a unit and how uncertain that classification is.

1.2 Bounded IRT and Bounded CDM

Feature	Bounded IRT	Bounded CDM
Latent representation	Continuous position θ_i	Discrete profile $\boldsymbol{\alpha}_i$
Typical output	Scores and uncertainty	Profile probabilities
Main purpose	Measurement along a scale	Domain-specific diagnosis

Both approaches can retain the bounded response scale (Noël and Dauvier 2007; Molenaar et al. 2022).

Binary domain states and the Q-matrix require substantive justification.

1.3 Foundations and Related Bounded CDMs

- Conjunctive cognitive assessment connects attribute profiles to item responses (Junker and Sijtsma 2001).
- G-DINA provides a broader framework for attribute interactions (Torre 2011).
- Earlier models extend diagnostic classification to continuous responses (Minchen et al. 2017; Oliveira et al. 2023).
- Jiang et al. (2025) develop a family of CDMs for bounded responses.
- The chapter develops the mean-precision B-DINA model of Oliveira et al. (2026a). Its comparison with the Jiang model is in the supporting information (Oliveira et al. 2026b).

2. Bounded Continuous Distributions: Beta Model

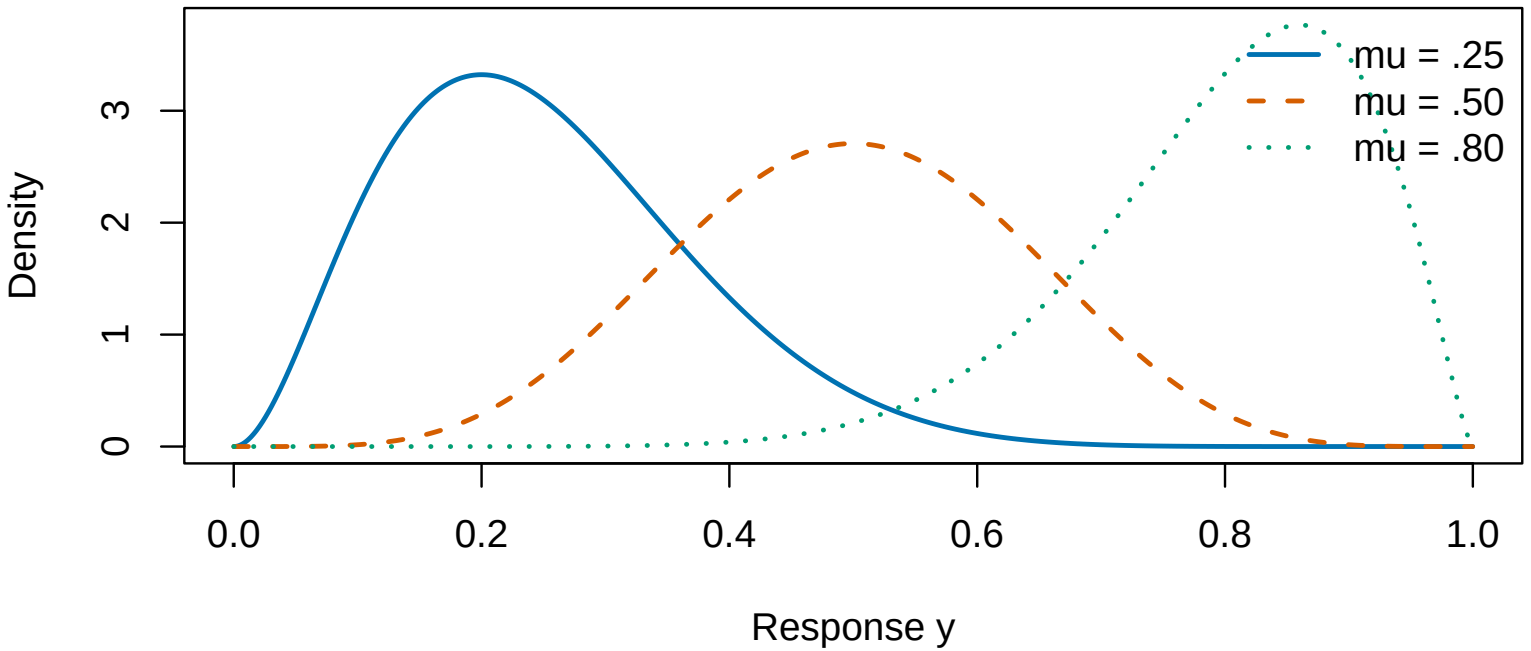
$$Y \sim \text{Beta}(\mu\phi, (1-\mu)\phi), \quad 0 < \mu < 1, \quad \phi > 0.$$

$$f(y \mid \mu, \phi) = \frac{y^{\mu\phi-1}(1-y)^{(1-\mu)\phi-1}}{B(\mu\phi, (1-\mu)\phi)}, \quad 0 < y < 1.$$

$$E(Y) = \mu, \qquad \text{Var}(Y) = \frac{\mu(1-\mu)}{1+\phi}.$$

The mean describes the expected coded indicator. Precision controls concentration at a fixed mean (Ferrari and Cribari-Neto 2004).

2.1 Beta Density Shapes



Common precision $\phi = 12$. Changing μ changes location and asymmetry.

2.2 Moment Restrictions and Endpoints

For any $Y \in [0,1]$ with mean m and variance v,

$$0 \leq v \leq m(1-m) \leq \frac{1}{4}.$$

For a beta distribution with finite positive parameters,

$$0 < v < m(1-m), \qquad \phi = \frac{m(1-m)}{v} - 1.$$

- A beta density can rise near a boundary while $P(Y=0) = P(Y=1) = 0$.
- Genuine endpoints need an explicit observation mechanism.
- Document any endpoint adjustment and assess its sensitivity.
- Rescale using theoretical bounds.

3. Cognitive Diagnosis: Attributes and Profiles

$$\boldsymbol{\alpha}_i = (\alpha_{i1}, \ldots, \alpha_{iK})^\top \in \{0,1\}^K, \qquad C = 2^K.$$

- $\alpha_{ik} = 1$ denotes possession of attribute k.
- The municipal attribute order is Economy, Health, Education.
- With $K = 3$, eight profiles are possible.
- Profile $(1, 0, 1)$ represents favorable Economy and Education conditions with unfavorable Health conditions.

These labels describe model-based latent conditions, rather than externally validated thresholds.

3.1 Q-Matrix and the Conjunctive Gate

$Q = (q_{jk})$ has J item rows and K attribute columns.

$$\eta_{ij} = \prod_{k:q_{jk}=1} \alpha_{ik}.$$

For $\mathbf{q}_j = (1, 0, 1)$:

Profile	Gate η_{ij}
$(1, 0, 1)$ or $(1, 1, 1)$	1
Any profile lacking Economy or Education	0

All required attributes must be present for the mastery regime (Junker and Sijtsma 2001). All-zero Q rows provide no learnable nonmastery regime.

3.2 Q-Matrix Construction and Validation

- Define attributes and inspect each indicator's substantive meaning.
- Document single- and multiple-attribute assignments.
- Assess whether joint requirements support a conjunctive interpretation.
- Compare plausible Q revisions using fit and classification stability.

Binary-response Q-validation methods provide useful precedents (Torre 2008; Torre and Chiu 2016). Their use with beta responses requires an appropriate adaptation.

Q specifies measurement relationships. It does not establish causal effects.

3.3 Beta B-DINA: Two Item Regimes

For item j (Oliveira et al. 2026a),

$$Y_{ij} \mid \boldsymbol{\alpha}_i, \boldsymbol{\Omega}_j, Q \sim \text{Beta}(\mu_{ij}\phi_{ij}, (1-\mu_{ij})\phi_{ij}),$$

$$\mu_{ij} = (1-\eta_{ij})\mu_{j0} + \eta_{ij}\mu_{j1}, \qquad \phi_{ij} = (1-\eta_{ij})\phi_{j0} + \eta_{ij}\phi_{j1}.$$

$$0 < \mu_{j0} \leq \mu_{j1} < 1, \qquad \phi_{j0}, \phi_{j1} > 0.$$

Responses are independent across items conditional on the profile and parameters. Each item has four response parameters.

3.4 Population Profiles and the Marginal Likelihood

Let $Z_i = c$ select profile $\boldsymbol{\alpha}^{(c)}$, and define

$$Z_i \mid \boldsymbol{\pi} \sim \text{Categorical}(\pi_1, \ldots, \pi_C), \quad \sum_c \pi_c = 1.$$

With f_{jr} the beta density in regime r,

$$p(\mathbf{y}_i \mid \boldsymbol{\Omega}, \boldsymbol{\pi}, Q) = \sum_{c=1}^{C} \pi_c \prod_{j=1}^{J} f_{j\eta_{cj}}(y_{ij}).$$

$$L(\boldsymbol{\Omega}, \boldsymbol{\pi}; Y, Q) = \prod_{i=1}^{N} \left\{ \sum_c \pi_c \prod_j f_{j\eta_{cj}}(y_{ij}) \right\}.$$

Population proportions π_c differ from individual posterior probabilities.

3.5 Posterior Classification

At posterior draw s, normalize the weighted profile likelihoods:

$$r_{ic}^{(s)} = \frac{\pi_c^{(s)} \prod_j f_{j\eta_{cj}}(y_{ij} \mid \boldsymbol{\Omega}_j^{(s)})}{\sum_d \pi_d^{(s)} \prod_j f_{j\eta_{dj}}(y_{ij} \mid \boldsymbol{\Omega}_j^{(s)})}.$$

$$P(Z_i = c \mid Y, Q) \approx \frac{1}{S} \sum_{s=1}^{S} r_{ic}^{(s)}.$$

- A modal profile summarizes classification.
- Attribute probabilities sum over profiles containing that attribute.
- Report uncertainty, especially when several profiles have similar support.

3.6 Identification and Interpretation

Binary DINA identification results motivate Q design (Gu and Xu 2019):

- Include an identity block I_K.
- Require each attribute in at least three items.
- Keep columns of the remaining Q^* distinct.

These conditions do not by themselves prove beta B-DINA identification.

Identical gate vectors make profiles observationally indistinguishable. Overlapping regimes and rare profiles can weaken finite-sample inference.

Joint identification of an unknown Q is a separate problem (Gu and Xu 2021). Mean ordering anchors regimes but need not imply stochastic dominance when precisions differ.

3.7 Item Parameters and Mean Contrasts

Binary DINA uses g_j and $1 - s_j$ as success probabilities. Beta B-DINA uses expected continuous responses μ_{j0} and μ_{j1}.

Oliveira et al. (2026a) define item difficulty as

$$\lambda_j = \mu_{j1} - \mu_{j0}.$$

- This is a mean contrast between attribute regimes.
- For proportions, a gap of 0.30 is 30 percentage points.
- The original-scale contrast is $(U - L)\lambda_j$.
- The contrast does not estimate a causal intervention effect.

Precisions also affect separation of the response distributions.

3.8 Distributional Discrimination

$$KS_j = \sup_{0 \le y \le 1} |F_{j1}(y) - F_{j0}(y)|.$$

For $a_{jr} = \mu_{jr}\phi_{jr}$ and $b_{jr} = (1 - \mu_{jr})\phi_{jr}$,

$$BC_j = \frac{B((a_{j0} + a_{j1})/2, (b_{j0} + b_{j1})/2)}{\sqrt{B(a_{j0}, b_{j0})B(a_{j1}, b_{j1})}},$$

$$H_j^2 = 1 - BC_j, \qquad H_j = \sqrt{1 - BC_j}.$$

Larger KS, H^2, or H means greater separation. Affinity BC decreases with separation.

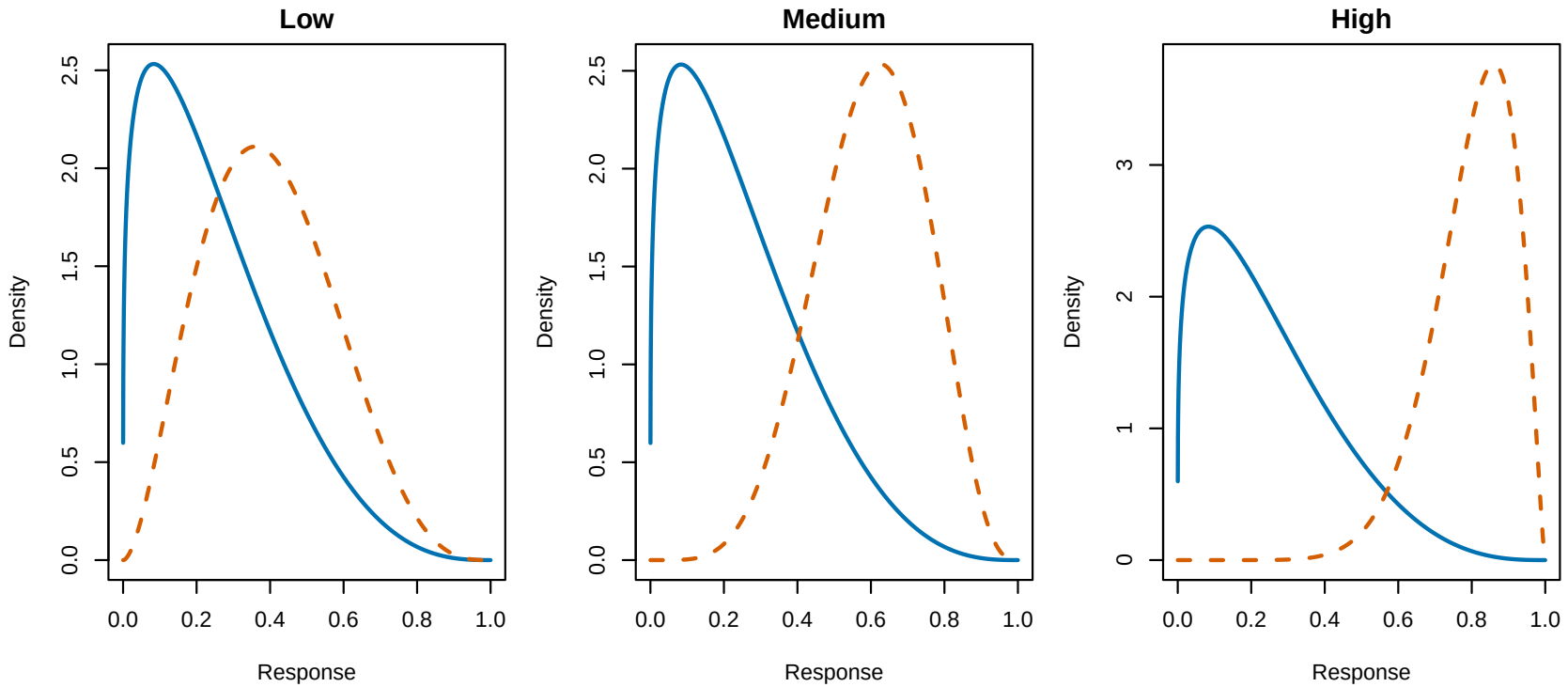
The integral labeled Hellinger in Oliveira et al. (2026a) is affinity; their simulation values are consistent with H^2. The chapter uses the distance conventions above explicitly.

3.9 Illustrative Item Configurations

level	mu0	mu1	phi0	phi1	KS	H2
Low	0.25	0.4	5	7	0.355	0.108
Medium	0.25	0.6	5	10	0.701	0.423
High	0.25	0.8	5	12	0.902	0.745

Configurations follow the simulation settings of Oliveira et al. (2026a). Means and precisions change together; these are not universal discrimination cutoffs.

3.10 Regime Density Separation



Nonmastery: solid blue. Mastery: dashed orange.

Classification also depends on Q, the other items, and profile proportions.

4. Bayesian Estimation: Prior Specification

$$\boldsymbol{\pi} \sim \text{Dirichlet}(1, \ldots, 1).$$

$$U_{j1}, U_{j2} \overset{\text{iid}}{\sim} U(0,1), \qquad \mu_{j0} = \min(U_{j1}, U_{j2}), \quad \mu_{j1} = \max(U_{j1}, U_{j2}).$$

$$\phi_{jr} \sim \text{Gamma}(0.01, 0.01), \quad r = 0, 1.$$

The Gamma parameters are shape and rate. These are the chapter's Model A priors (Oliveira et al. 2026a; Bazán et al. 2026). Q remains fixed.

4.1 Prior Implications

Sorting auxiliary uniform means induces

$$\mu_{j0} \sim \text{Beta}(1, 2), \quad \mu_{j1} \sim \text{Beta}(2, 1), \quad p(\lambda_j) = 2(1 - \lambda_j).$$

- The ordered means are dependent, with expected gap $1/3$.
- Gamma$(0.01, 0.01)$ has mean 1, variance 100, and high density near zero.
- Large prior variance alone does not establish weak information.
- Prior predictive checks and sensitivity analyses should examine regime overlap, precision, and profile proportions.

4.2 Posterior and MCMC Updates

$$p(\boldsymbol{\Omega}, \boldsymbol{\pi}, \mathbf{Z} \mid Y, Q) \propto p(\boldsymbol{\Omega}, \boldsymbol{\pi}) \prod_i \left\{ \pi_{Z_i} \prod_j f_{j\eta_{Z_i,j}}(Y_{ij}) \right\}.$$

$$P(Z_i = c \mid \cdots) \propto \pi_c \prod_j f_{j\eta_{cj}}(Y_{ij}),$$

$$\boldsymbol{\pi} \mid \mathbf{Z}, \cdots \sim \text{Dirichlet}(1 + n_1, \ldots, 1 + n_C).$$

Item-parameter updates are nonconjugate. JAGS assigns samplers to the graph's stochastic nodes. The gate and ordered means are deterministic (Plummer 2017; Neal 2003).

4.3 JAGS Response and Parameter Blocks

```
for (i in 1:N) {
  Z[i] ~ dcat(pai[])
  for (j in 1:J) {
    eta[i,j] <- gate[Z[i],j]
    mu[i,j] <- (1-eta[i,j])*u0[j] + eta[i,j]*u1[j]
    phi[i,j] <- (1-eta[i,j])*phi0[j] + eta[i,j]*phi1[j]
    Y[i,j] ~ dbeta(mu[i,j]*phi[i,j],
                   (1-mu[i,j])*phi[i,j])
  }
}
for (j in 1:J) {
  u[j,1] ~ dunif(0,1)
  u[j,2] ~ dunif(0,1)
  u0[j] <- min(u[j,1],u[j,2])
  u1[j] <- max(u[j,1],u[j,2])
  phi0[j] ~ dgamma(0.01,0.01)
  phi1[j] ~ dgamma(0.01,0.01)
}
pai[1:C] ~ ddirch(delta[])
```

Precompute `gate[c,j]` from Q and profiles. Set `delta = rep(1,C)`.

4.4 Sampling Settings and Convergence

Setting	Repository example	Main paper
Chains	2	2
Total iterations per chain	5,000	25,000
Burn-in per chain	1,000	5,000
Thinning	1	1
Pooled retained draws	8,000	40,000

Sources: main paper (Oliveira et al. 2026a); supplementary comparison and repository example (Oliveira et al. 2026b; Bazán et al. 2026).

In `R2jags`, `n.iter` includes burn-in; adaptation is separate (Su and Yajima 2024).

Inspect trace plots, between-chain agreement, effective sample sizes, and Monte Carlo error. Retained draws remain autocorrelated. Run length alone cannot establish convergence.

4.5 Posterior Predictive Checks

Oliveira et al. (2026a) use the total-response statistic

$$T(Y) = \sum_{i,j} Y_{ij}, \qquad PPP = P\{T(Y^{rep}) \geq T(Y) \mid Y\}.$$

- A value near 0.5 indicates agreement for this statistic.
- Examine item variances, quantiles, and residual dependence as well.
- Check whether the model reproduces indicator shapes and regime contrasts.
- A satisfactory total-response check does not validate every model assumption.

4.6 Predictive Comparison and Likelihood Targets

For a new unit, use the person-marginal log likelihood:

$$\ell_i^{(s)} = \log \left\{ \sum_c \pi_c^{(s)} \prod_j f_{j\eta_{cj}}(y_{ij}) \right\}.$$

- A conditional item likelihood targets a different prediction task.
- Use a draws-by-unit matrix for WAIC or PSIS-LOO.
- Use the variance penalty for `loo::waic` and inspect Pareto-k diagnostics for PSIS-LOO (Vehtari et al. 2017).
- Continuous densities can exceed one, allowing negative deviance values.

Compare criteria only when their likelihood target and conventions agree.

4.7 WAIC and DIC: State the Convention

For the same likelihood target, define

$$\mathrm{lppd} = \sum_i \log \left\{ S^{-1} \sum_s e^{\ell_i^{(s)}} \right\}, \qquad \mathrm{WAIC} = -2(\mathrm{lppd} - p_{\mathrm{WAIC}}).$$

- `loo::waic` uses $p_{\mathrm{WAIC}} = \sum_i \mathrm{Var}_s(\ell_i^{(s)})$ (Vehtari et al. 2017).
- The supporting information prints the alternative penalty

$$2 \sum_i \left[\log E_s\{e^{\ell_i^{(s)}}\} - E_s(\ell_i^{(s)}) \right]$$

  and reports $\mathrm{DIC} = \bar{D} + \mathrm{Var}(D)/2$ (Oliveira et al. 2026b).
- For unreliable PSIS-LOO, refit or use K-fold validation. Preserve chain information when estimating relative efficiency.

4.8 Model A, Model B, and Computation

- Model A uses ordered beta means and regime-specific precisions.
- The supplementary Model B uses a log-shape parameterization motivated by Jiang et al. (2025; Oliveira et al. 2026b; Bazán et al. 2026).
- Compare the induced priors and mean ordering before interpreting model comparisons.
- Stan requires marginalizing discrete profile assignments; posterior profile probabilities can then be recovered from continuous draws (Stan Development Team 2026).
- Alternative implementations should be compared using accuracy, diagnostics, and runtime.

5. Municipal Application: Data and Coding

- The supplied snapshot contains 1,668 municipalities and 19 indicators.
- Attributes are Economy, Health, Education.
- Responses are already coded toward favorable conditions, including $1 - \mathrm{GINI}$ for equality.
- All supplied responses lie strictly inside $(0, 1)$.

The main paper's introductory count of 1,688 differs from the 1,668 observed rows. The supplementary comparison also uses 1,668 (Oliveira et al. 2026a, 2026b; Bazán et al. 2026).

Municipal indicators are items. Municipalities are observational units.

5.1 Municipal Q-Matrix: Items 1–10

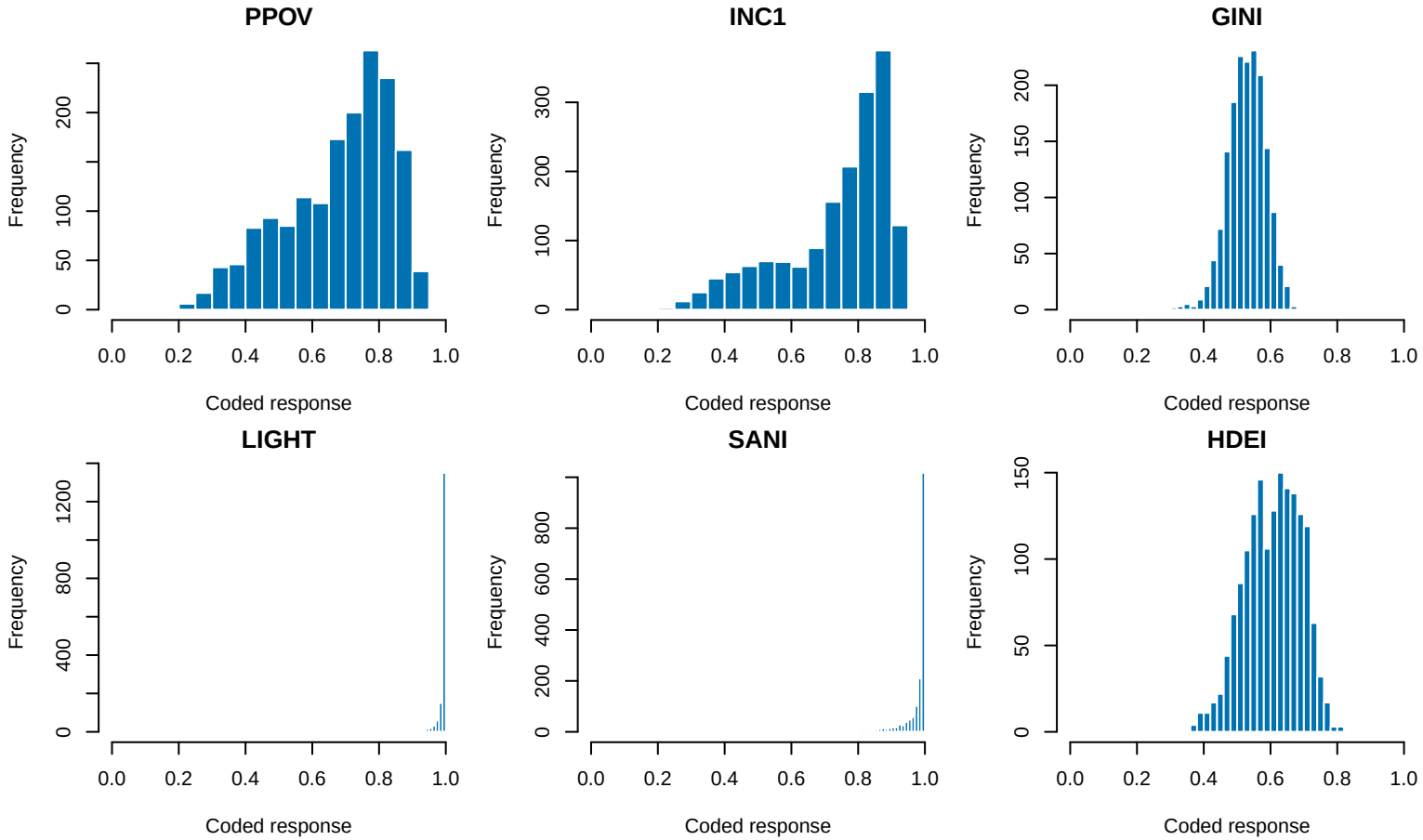
Item	Economy	Health	Education
DEP	1	0	0
S60	0	1	0
ILIT	1	0	1
HIGH	1	0	1
GINI	1	0	0
PPOV	1	0	0
INC1	1	0	0
UNEM	1	0	0
BWAT	1	1	0
GARB	1	1	0

5.2 Municipal Q-Matrix: Items 11–19

Item	Economy	Health	Education
LIGHT	1	0	0
SANI	0	1	0
ELPOV	1	0	1
CHILD	0	1	1
HEAD	0	0	1
WORKS	1	0	1
HDEI	0	0	1
HDLI	0	1	0
HDII	1	0	0

Water and garbage collection require Economy and Health jointly. These assignments encode substantive measurement hypotheses.

5.3 Observed Indicator Distributions



5.4 Descriptive Results and Conditional Modeling

Indicator	Mean	SD
LIGHT	0.990	0.017
SANI	0.976	0.040
HIGH	0.287	0.092
HDEI	0.608	0.083

PPOV and INC1 have SDs near 0.160 and 0.161.

Marginal distributions mix over profiles. They motivate a flexible bounded model but cannot establish conditional beta fit, local independence, or the DINA gate.

5.5 Saved Repository Fit: Sampling Diagnostics

Parameter	Rhat	n.eff
phi1[11]	6.501	2
phi0[6]	1.204	12
phi0[19]	1.145	15
u0[6]	1.121	17
u0[13]	1.090	22
u0[4]	1.085	23

Selected parameters with the largest saved R-hat values. Poor between-chain agreement limits interpretation of this run. Longer sampling may help, but requires renewed diagnostic checks.

5.6 Published Item Means and Contrasts

item	mu0	mu1	mean_gap
PPOV	0.500	0.765	0.265
INC1	0.606	0.841	0.235
CHILD	0.938	0.940	0.002
GINI	0.521	0.543	0.022
UNEM	0.933	0.943	0.010
LIGHT	0.979	0.995	0.016
SANI	0.958	0.995	0.037

Mean gaps use rounded published estimates. Precision differences can separate regimes even when means are close (Oliveira et al. 2026a).

5.7 Published Precision and Discrimination

item	sqrt_phi0	sqrt_phi1	KS_published	H_published
PPOV	3.939	4.727	0.78	0.53
INC1	3.048	6.701	0.75	0.49
CHILD	5.533	6.108	0.02	0.00
GINI	9.889	8.739	0.17	0.03
UNEM	7.867	8.924	0.12	0.01
LIGHT	7.505	190.939	0.80	0.57
SANI	4.756	123.062	0.85	0.57

Oliveira et al. (2026a) report $\sqrt{\phi}$, which must be squared in the beta variance formula. The H column retains the source label; its convention differs from the distance H defined earlier.

5.8 Published Population Profile Proportions

Economy	Health	Education	proportion
0	0	0	0.279
1	0	0	0.059
0	1	0	0.029
0	0	1	0.010
1	1	0	0.055
1	0	1	0.157
0	1	1	0.004
1	1	1	0.407

Table 10 of Oliveira et al. (2026a). Attribute order: Economy, Health, Education. Population proportions are distinct from individual diagnoses.

5.9 Repository and Published Summaries

Quantity	Published	Saved_repository
All-positive proportion	0.407	0.419
Total-response PPP	0.404	0.372

Published values are from Oliveira et al. (2026a); saved summaries are from Bazán et al. (2026). The shorter run has diagnostics that require attention. Similar point estimates do not establish equivalent posterior inference.

The published all-positive profile interval is transcribed as [.407, .432] in the chapter. Consult the source before interpreting that unusual lower limit.

6. Exercises: Estimation and Measurement

1. Increase MCMC run lengths and burn-in. Compare effective sample sizes, Monte Carlo error, and profile probabilities.
2. Fit bounded IRT models to the municipal data. Compare the information in continuous scores and discrete attribute profiles.
3. Fit B-DINA to the Application 2 visual analogue responses analyzed by Jiang et al. (2025) and supplied by Bazán et al. (2026). Document coding, attributes, Q, and classification uncertainty.
4. Assess prior sensitivity and compare substantive conclusions across plausible Q specifications.

6.1 Exercises: Parameterizations and Computation

1. Express repository Models A and B using beta means and precisions. Compare priors and ordering constraints. Does Model B guarantee a higher mastery mean?
2. Reproduce the Models A/B comparisons for Applications 1 and 2 in Oliveira et al. (2026b). State the likelihood target and fit-criterion convention.
3. Investigate faster estimation through marginalization over profiles or alternative MCMC algorithms. Compare accuracy as well as runtime.

Use the local repository snapshot and the references in the companion chapter (Bazán et al. 2026).

7. Conclusions

- Beta B-DINA connects continuous bounded indicators to discrete attribute profiles.
- Q and the conjunctive gate define how attributes enter each item.
- Regime means and precisions jointly determine response separation.
- Posterior profile probabilities retain classification uncertainty.
- Credible diagnosis requires substantive attribute definitions, a defensible Q, adequate sampling, and model checks.
- Municipal classifications describe latent domain conditions and support further investigation.

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